brückenbauer GmbH

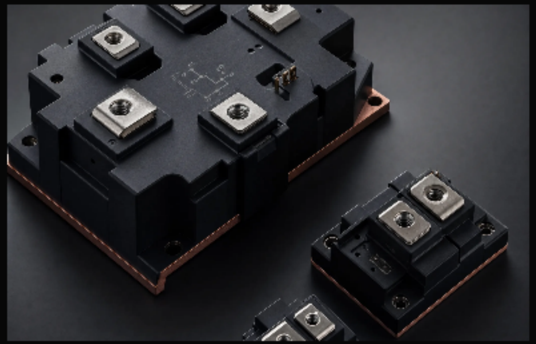
Power Electronics Portfolio

2026

Premium electronic component portfolio for industrial innovation.
Carefully selected technologies from trusted global partners.

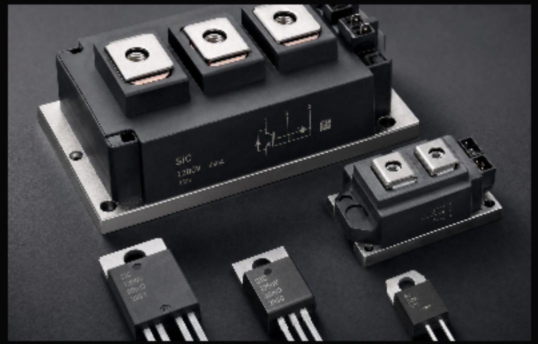
Four power families define the portfolio core.

The source portfolio centers on semiconductor switching and regulated conversion. This structure gives buyers a clean path from operating conditions to component family.



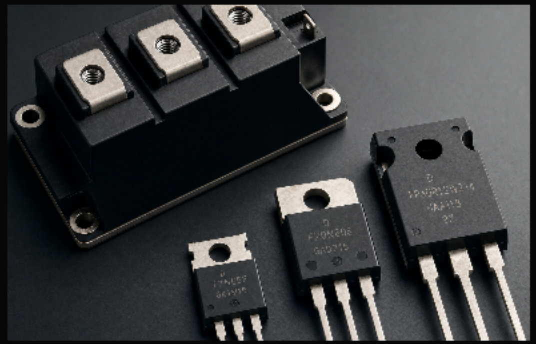
IGBT

High-current bridge and module platforms



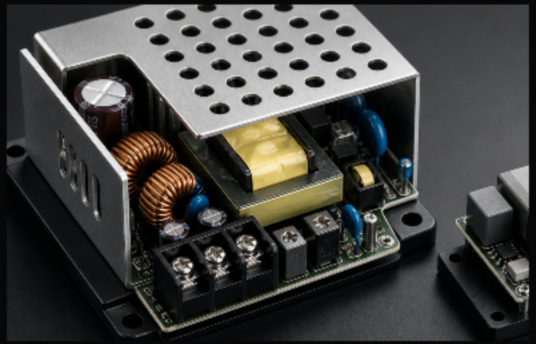
SiC MOSFET

High-efficiency switching for compact power designs



MOSFET

Discrete, module and diode options for scalable circuits



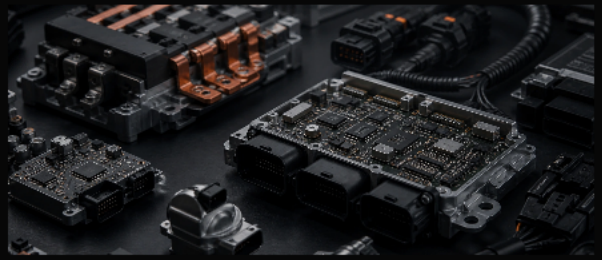
Converters

AC/DC and DC/DC power supply platforms

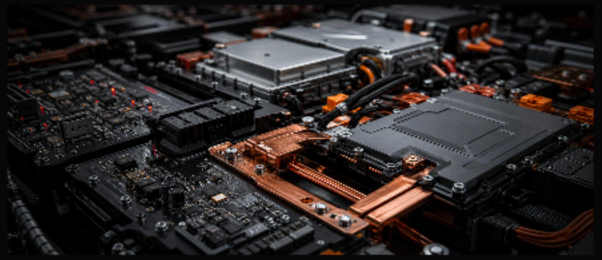
Power electronics serves demanding environments.

The source document maps power management into aerospace, industrial, automotive, medical, building, HVAC and renewable energy contexts.

Industrial automation



Automotive and transport



Renewable energy

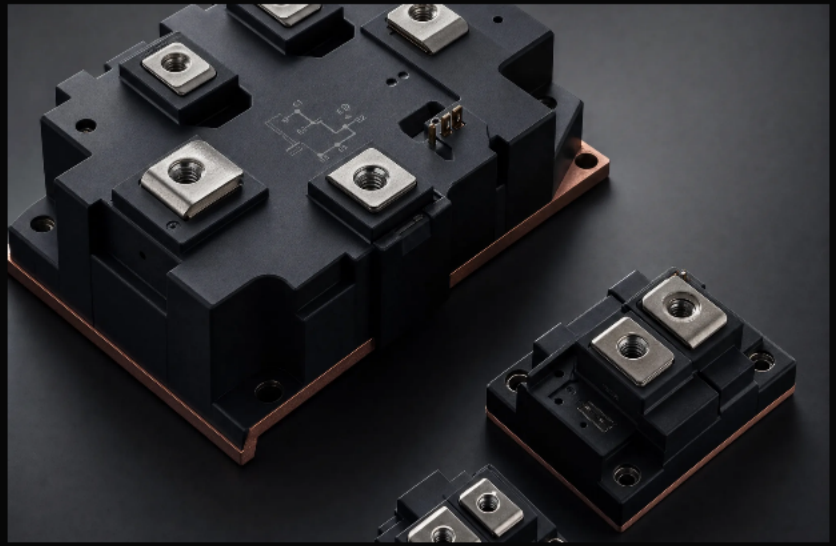
Grid infrastructure

Medical systems

Aerospace and defense

IGBT modules support high-current bridge applications.

IGBT families cover 650 V and 1200 V classes with current ratings from 10 A to 600 A, where density, reliability and controlled losses matter.



Performance parameters

Voltage classes	650 V and 1200 V
Current ratings	10 A to 600 A
Power loss	Low switching losses for efficiency
Reliability	Materials selected for service life

Key fit

Half bridge

3-phase bridge

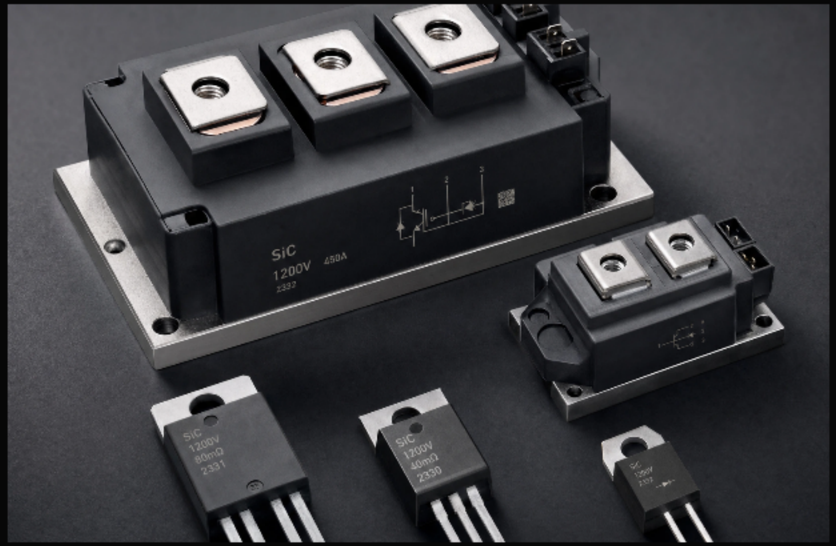
SiC module routes

Target applications

Industrial drives / Automotive / Renewable energy / Home appliances

SiC MOSFETs enable smaller, faster and cooler designs.

Silicon carbide options reduce conduction and switching losses, making them useful where thermal performance and high-frequency operation drive the design.



Performance parameters

On-resistance	Lower conduction losses
Switching frequency	Supports smaller and lighter designs
Breakdown voltage	Safe operation under demanding loads
Temperature	Suitable for harsh operating conditions

Key fit

Half bridge

Power diode

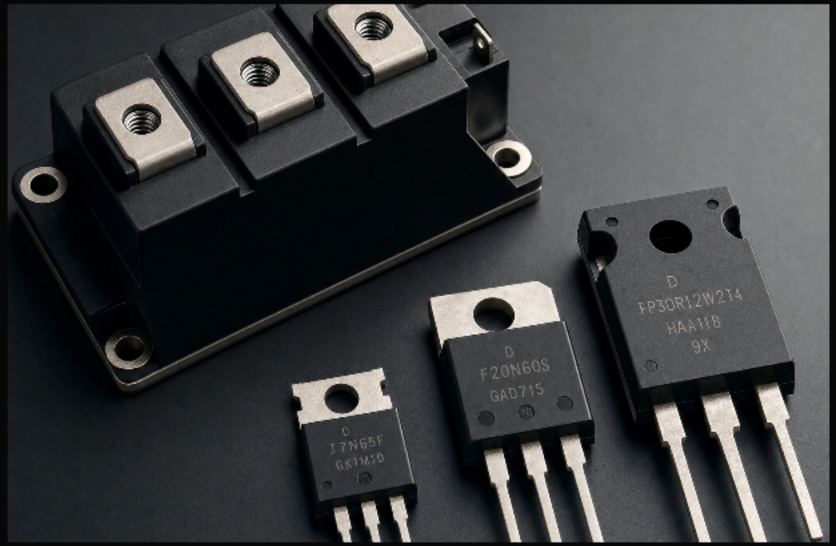
3-phase routes

Target applications

E-mobility / Industrial automation / Renewable energy / Power electronics

MOSFETs keep low and medium power switching scalable.

Discrete, bare-die, module and diode options support fast switching, low drive power and efficient operation across compact power circuits.



Performance parameters

Voltage range	20 V to 45 V discrete options
Packages	TO-220, TO-252 and model-specific formats
Bare-die range	30 V to 45 V
Design fit	Compact layouts for constrained assemblies

Key fit

Rectifiers

MOSFET modules

Power diodes

Target applications

Drive technology / Welding / Wind turbines / Photovoltaic systems

Converters translate stable power into usable system behavior.

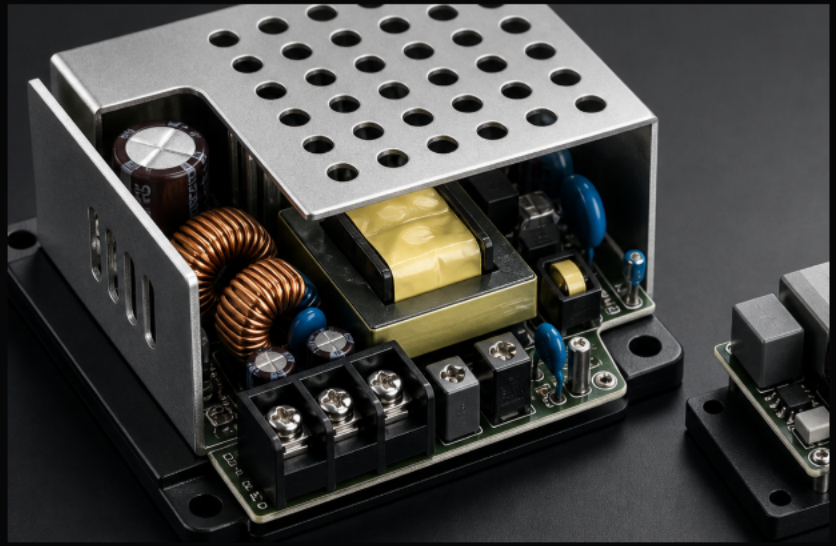
AC/DC, DC/DC, power supply and plug-converter platforms are selected around output accuracy, efficiency, ripple, overload and short-circuit protection.

Performance parameters

Output accuracy	Stable output across input and load range
Efficiency	High energy conversion efficiency
Ripple and noise	Low-noise output for sensitive loads
Protection	Short-circuit and overload handling

Target applications

PV energy storage / Railway / Industrial control / Medical / LED lighting



Key fit

AC/DC

DC/DC

Power supply

Plug converters

The right family depends on the operating envelope.

Instead of forcing a part too early, the portfolio should first qualify voltage, current, frequency, thermal behavior, package constraints and compliance requirements.

High current bridge	IGBT	Industrial drives, renewable energy, home appliances
High frequency and temperature	SiC MOSFET	E-mobility, compact conversion, demanding thermal environments
Low to medium power switching	MOSFET	Drives, welding, photovoltaics, general power electronics
Regulated supply and protection	Converters	Railway, ICT, medical, LED and energy storage systems

Qualification needs technical, compliance and lifecycle evidence.

The source document references IEC, JEDEC, UL and EMC test frameworks. Brueckenbauer can keep those checks connected to sourcing, sample planning and long-term availability.

Electrical envelope	Voltage class, current rating, losses and switching behavior.
Thermal reliability	Operating temperature, thermal cycling and package behavior.
Compliance fit	IEC 60747, JEDEC, UL 62368 and EMC expectations where applicable.
Lifecycle planning	Availability, alternates, samples and obsolescence risk before commitment.

Qualified suppliers expand sourcing options.

Brückenbauer works with a network of qualified specialist manufacturers covering power modules, semiconductors, converters, cooling solutions and related power electronics. Our value lies in identifying the most suitable sourcing route for each technical and commercial requirement.

AIPUPOWER Source-listed partner	GeneSiC Source-listed partner	CLNF Source-listed partner	Mitsubishi Electric Source-listed partner	alfatec Source-listed partner
Yuan Dean Source-listed partner	Silvermicro Source-listed partner	SemiQ Source-listed partner	TECHSEM Source-listed partner	FANSO Source-listed partner

RFQ PATHWAY

Move from operating conditions to a qualified RFQ.

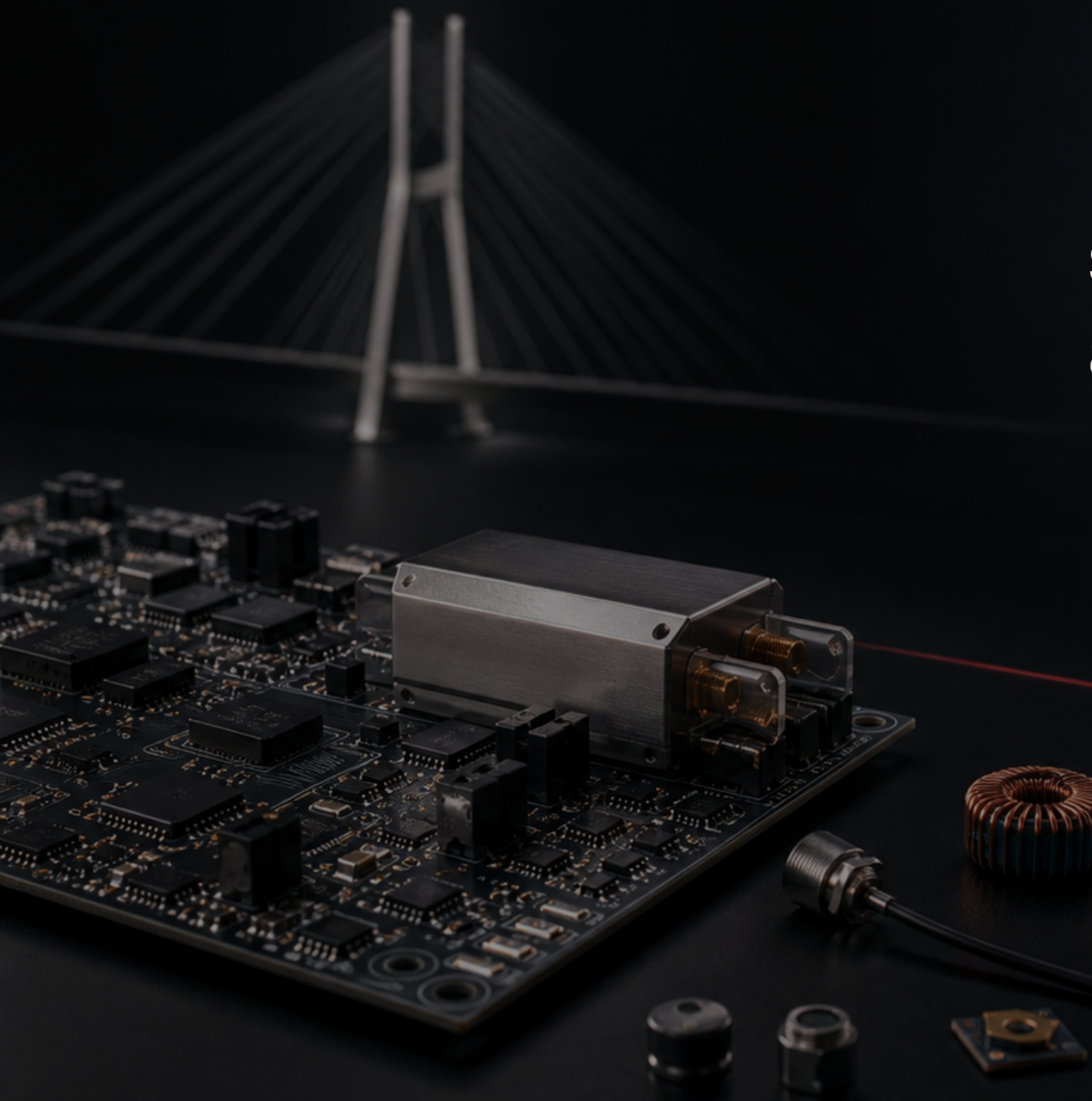
Our portfolio is organized into a customer-focused structure that simplifies technical evaluation, product selection, and RFQ preparation.

01 Define voltage, current and switching profile

02 Choose the most likely component family

03 Check package, thermal and compliance constraints

04 Prepare RFQ, samples and alternate routes



Power portfolio review starts with the application.

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